

developed extremely lightweight VR glasses—the AjnaX Glasses—which weigh just around 150gm compared to the average 500gm that popular international brands weigh. This makes it possible for trainees to comfortably wear the glasses and train for long durations. This initiative by the Karnataka government aims to upskill more than 9000 students this year. AjnaLens also provides end-to-end enterprise VR solutions, digital twin solutions, and indigenous defence solutions.

Collaboration and productivity: Work smarter

Companies are betting big on the metaverse to help employees collaborate better, unmindful of geographical distances. Microsoft Mesh, Workplace from Meta (earlier Facebook), Gather, Teamflow, and Virbela are some of the key players in this space. Instead of video-calling colleagues and collaborating using ineffective tools like screen-sharing, the metaverse will enable your *avatars* to gather in a virtual meeting room and collaborate. In fact, you can probably do things more smartly than in a physical office, because the virtual space would enable you to pull holographic images out of your hat, to explain things or make changes to a prototype!

The metaverse is especially handy in a factory environment, where technicians are using both hands and cannot simultaneously use the laptop. Imagine a mechanic working on something complex. What if he had a headset like Microsoft's HoloLens to access the schematics and instructions needed to complete the repair? Would he not save so much time? Also imagine, when a mechanic is repairing something and needs an expert's opinion, how would it be if the expert could see the object being repaired through the eyes of the mechanic and explain things in real-time?



Figure 3: Popular martial arts workout BodyCombat combines fitness with gaming in the metaverse (Courtesy: Les Mills)

How BMW has put all this to work

More than a year ago, BMW Group deployed one of the early examples of the metaverse, when the term was still not so well-known. They deployed an augmented reality (AR) application for vehicle concept and prototype engineering at their pilot plant in Munich. Vehicles and their components are visualised on a platform linked to the BMW Group's product data management system. Specialists can drag and drop CAD files of components from the web based database to the AR goggles, to reproduce the data in 3D and in their original size in a realistic environment. Users can modify the size and position of components with simple hand movements, literally playing around with the design. People at different locations around the world can team up to review designs and concepts together. This helps identify errors much earlier in the process, saving cost and time later. The company said in a press release that the AR application speeded up the process by around 12 months!

Last November, BMW announced the adoption of Nvidia's Omniverse to simulate every aspect of its manufacturing. The idea is for each of its 31 factories to have a fully-

functioning, real-time digital twin capable of simulating production and finite scheduling, right down to work order instructions and robotics programming on the shop floor. This will help in improving product quality and output, while reducing manufacturing costs and unplanned downtime. It will enable global teams to collaborate on designing and planning the factory in real time. With the digital twin of an entire factory, BMW engineers can quickly identify where and how each specific model's production sequence can be improved. It also gives BMW the flexibility to reconfigure its factories quickly to accommodate new model launches.

The company has also armed technicians with RealWear HMT-1 headsets, running TeamViewer, in 347 BMW centres across the US to reduce vehicle service times. Using voice commands, technicians can access and view technical documents on a high-resolution micro-display or remotely connect with BMW engineers for expert advice, all the time keeping their hands free to work! HMT-1's camera enables offsite engineers to see exactly what's in front of the local technician, so they can help them troubleshoot the issue in real time. The offsite expert can also markup live video feeds or image stills, which the mechanic can